

Active Video Perception: Iterative Evidence Seeking for Agentic Long Video Understanding

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<https://activevideoperception.github.io/>

Abstract

*Long video understanding (LVU) is challenging because answering real-world queries often depends on sparse, temporally dispersed cues buried in hours of mostly redundant and irrelevant content. While agentic pipelines improve video reasoning capabilities, prevailing frameworks rely on a query-agnostic captioner to perceive video information, which wastes computation on irrelevant content and blurs fine-grained temporal and spatial information. Motivated by active perception theory, we argue that LVU agents should actively decide what, when, and where to observe, and continuously assess whether the current observation is sufficient to answer the query. We present **Active Video Perception (AVP)**, an evidence-seeking framework that treats the video as an interactive environment and acquires compact, query-relevant evidence directly from pixels. Concretely, AVP runs an iterative plan–observe–reflect process with MLLM agents. In each round, a planner proposes targeted video interactions, an observer executes them to extract time-stamped evidence, and a reflector evaluates the sufficiency of the evidence for the query, either halting with an answer or triggering further observation. Across five LVU benchmarks, AVP achieves highest overall accuracy with significant improvements. Notably, AVP outperforms the best agentic method by 5.7% in average accuracy while only requires 18.4% inference time and 12.4% input tokens.*

1. Introduction

From streaming platforms to TV programs, video has become a primary medium for capturing and conveying information. However, long video understanding (LVU) remains challenging because it demands the ability to localize and

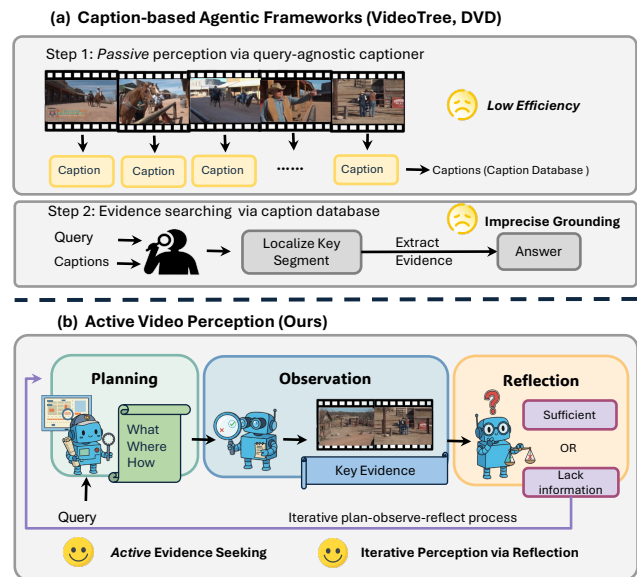


Figure 1. **Motivation of Active Video Perception.** Prior methods follow a *passive* perception paradigm which leverage query-agnostic captioner to perceive the video information, leading to low efficiency and imprecise visual grounding. Instead, we *actively* perceive query-relevant content by treating the long video as an interactive environment to be explored in a goal-directed manner.

integrate sparse, temporally dispersed cues across long time spans. Although recent multimodal large language models (MLLMs) [3, 23–25, 36, 53, 55, 61] substantially improve visual recognition, naively applying them to densely sampled, full-length videos is both computationally costly and brittle for complex queries: most video tokens are redundant, while the brief, localized evidence that actually matters is diluted or overlooked in the long sequence.

These limitations have motivated a recent surge of agentic approaches for long video understanding [62, 65, 75, 78]. Rather than treating the video as a single monolithic input,

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these methods use LLMs to orchestrate perception and reasoning over the video through planning. However, recent leading methods [46, 82, 88] still rely on captioners to convert visual information into text space as the primary interface for LLM reasoning and tool calling. This caption-based framework leverages LLMs’ strengths in text processing but introduces two inherent limitations:

1. **High Computational Cost:** Query-agnostic captioning generates large amounts of irrelevant information, expending computation on unrelated content and resulting in low efficiency.
2. **Imprecise Grounding via Captions:** Existing approaches use captions to localize key events, which may discard fine-grained temporal and spatial cues and weaken causal tracing.

These limitations underscore the need for an agentic framework that adaptively focuses on informative video regions, seeks query-related evidence directly over video pixels while maintaining high efficiency.

We take inspiration from how humans inspect long videos: we do not need to watch every frame; instead, we plan our observation based on the query. For instance, given a question about a specific plot, we first skim the video for coarse cues (plot localization), then take targeted observation by focusing on the key video regions for detail clues. Active perception theory [1, 4, 5] formalizes this behavior: *“An agent is an active perceiver if it knows why it wishes to sense, and then chooses what to perceive, and determines how, when and where to achieve that perception”*. Even though active perception concept is mainly used in robotics domain [44, 50, 69], We argue that agentic LVU frameworks can similarly benefit from query-driven, temporally grounded observation that decides what, when, and where to look, while continually assessing whether the accumulated evidence is sufficient for the query or whether further observation is required.

Building on this view, we propose **Active Video Perception (AVP)**, an agentic evidence seeking framework for long video understanding. As shown in Fig. 1, rather than passively perceiving the video by captioning, AVP treats the video as an interactive environment and actively decides what/where/how to observe the video to acquire the query-related information. This targeted observation design allows AVP to focus on the key informative segments, avoid redundant processing over static or irrelevant content, ultimately improving both efficiency and reliability on complex long-horizon queries.

Since complex queries often depend on sparse or ambiguous cues that cannot be resolved in a single round of observation, AVP adopts an iterative plan–observe–reflect process with MLLM agents. In each round, a planner proposes targeted interactions with the video by deciding what to inspect, where to focus, and at what granularity. Then, an observer

executes these plans to extract compact, time-stamped evidence. Finally, a reflector evaluates the query-sufficiency of the extracted evidence and decide whether additional round of observation is needed. If the extracted evidence is insufficient, it appends the current plan, evidence, and justification to the running history to guide the planner in deciding the next plan. This closed-loop process enables AVP to progressively refine its focus, revisit uncertain moments, and allocate computation adaptively, leading to more efficient processing and reliable reasoning on long, complex videos.

We demonstrate the effectiveness and efficiency of AVP by evaluating it on five long video understanding benchmarks, including MINERVA [34], LVBench [70], VideoMME [16], MLVU [84] and LongVideoBench [68]. Compared to the existing agentic approaches, AVP attains higher accuracy while using substantially less compute by formulating LVU as goal-conditioned observations. Specifically, compared to the leading agentic method DeepVideoDiscovery (DVD) [82], AVP achieves an average accuracy gain of **5.7%**. What’s more, on LVBench, AVP achieves better performance while only consuming **18.4%** inference time and **12.4%** input tokens compared to DVD, validating the efficiency of AVP. We further conduct extensive ablation studies that highlight and justify the key design choices of AVP.

2. Related Work

Long Video Understanding The advancement in long video understanding (LVU) benchmarks [7, 16, 68, 70, 84] has extended video reasoning problem from short clips to realistic scenarios, involving multi-minute or hour-long videos. To address this, previous video-specific MLLMs [27, 42, 43, 56, 58, 83] mainly focus on the challenge of excessive token inputs by extending the context length [9, 81], reducing the video tokens [26, 45, 49, 63] or keyframe selection [2, 6, 51, 59, 72, 74, 76, 87]. Notably, VAP [30] also introduce the concept of “action perception” to LVU task, they treats key frame selection as data acquisition in active perception and leverages a lightweight text-conditioned video generation model to represent prior world knowledge. Instead, AVP treats LVU as query-driven evidence seeking in video environments. As a result, AVP tackles complex LVU task by focus perception in key regions, achieves significantly better efficiency.

Recently, inspired by the great success of DeekSeek-R1 [10], several works [15, 57, 64, 67] explore the Chain-of-thoughts video reasoning model. Later works [17–19, 39, 52, 60, 77, 80] explore the idea of “Thinking with Video”, which incorporate visual CoT strategy to conduct coarse-to-fine video exploration. Compared to these methods, AVP has two clear advantages: (1) query-adaptive, previous work mainly follows a coarse-to-fine schema with fixed FPS/resolution setup, instead, AVP decides what/where/how

to observe the video based on the query; (2) training-free, instead of generating large-scale training samples with reasoning trace, we directly employ an agentic approach and significantly reduce compute cost.

Agentic Frameworks for Long Video Understanding

LLM-based agents that combine reasoning, planning, and action have been widely studied [12, 48, 73] in NLP domain, which also draws growing attention in LVU community. To decouple the complex LVU task, early agentic frameworks [13, 20, 21, 31, 41, 65, 66, 78, 79] adopt a captioner-LLM design: video segments are converted into captions, which an LLM then uses the generated caption to answer the video query. Meanwhile, several works [14, 22, 28, 32, 33, 47, 86] utilize the idea of “visual programming”, decompose the complex query into multiple steps to leverage expert modules. Reflection-based frameworks [8, 85] add a verification agent after the initial answering process to refine the reasoning. Building on these works, recent studies [8, 11, 29, 40, 46, 71, 82, 88] aim to improve evidence retrieval and reasoning efficiency in text space. Notably, VGenT [46] constructs a caption-based graph to enable long-range retrieval and relational reasoning across segments. VideoLucy [88] introduces a memory backtracking mechanism that allows the model to revisit earlier multi-scale text captions during multi-step reasoning. Deep Video Discovery [82] uses tool-based search to iteratively refine textual evidence over long videos. Instead of relying on captioners, AVP reasons directly over visual inputs through an iterative plan-observe-reflect process, selectively watching only what the query requires and maintaining a compact evidence record. This active, iterative video observation design preserves fine-grained grounding while avoiding the redundancy and overhead of caption-based LVU pipelines.

3. Method

We present **Active Video Perception (AVP)**, an iterative evidence seeking framework for agentic LVU. AVP is inspired by the concept of *active perception* [1, 4, 5], which argues “a complete artificial agent necessarily must include the ability of knowing why it wishes to sense, and then choosing what to perceive, and determining how, when and where to achieve that perception”. Through the lens of active perception, we formulate LVU task as query-driven evidence seeking in video environments, where the LVU agent iteratively decides what, where and how to interact with the video to find the key evidence based on previous observation.

Concretely, as shown in Fig. 2, given a query Q and a video V , AVP runs an iterative plan-observe-reflect process with MLLM agents. In each round, a planner first proposes observation plan by choosing what to inspect, where to focus, and how to sample. An observe agent executes that plan to extract compact, time-stamped evidence by observing the

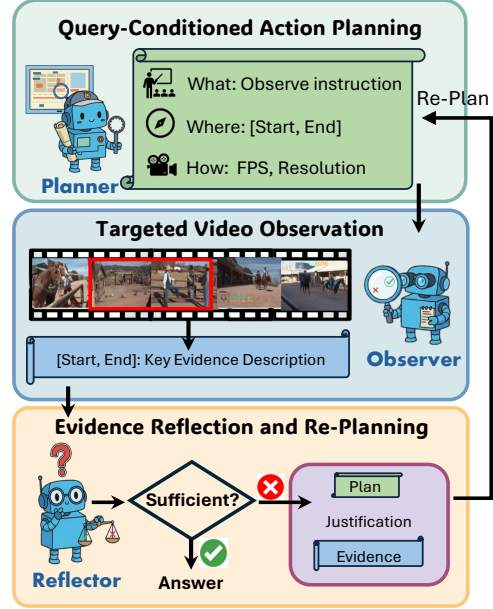


Figure 2. **Framework of Active Video Perception (AVP)**. AVP operates by an iterative plan-observe-reflect process with MLLM agents. At each round, the planner decide what/where/how to interact with the video, the observer extract structured query-related evidence by executing the plan and the reflector evaluates the extracted evidence to decide whether an additional round is need.

video purposefully. A reflector verifies evidence against the query to estimate the confidence; if it exceeds the confidence threshold, AVP outputs the answer and stops, otherwise it returns a justification to guide the next round of observation planning. We iterate this process until either a sufficiently confident answer is obtained or the round limit is reached. We introduce each component in detail as follow.

3.1. Query-Conditioned Action Planning

Inspired by active perception concept, instead of passively processing frames uniformly or converting into caption list, AVP first plans deciding what, where, and how to observe the long video to obtain the query-related evidence. Specifically, AVP leverage a planner (PLANNER) to decide what to look for, where to look, and how to observe in order to solve the give query.

Initial Plan. At round $r=1$, given query Q and video V , the PLANNER instantiates a concrete observation specification that states what to observe, the region to inspect, and how to sample. We initialize

$$P^{(1)} \leftarrow \text{PLANNER.INIT}(Q),$$

and represent it as

$$P^{(1)} = (\text{what}^{(1)}, \text{where}^{(1)}, \text{how}^{(1)}).$$

- **what** is a brief, query-conditioned instruction naming the key evidence to seek (e.g., “locate the moment the coach enters,” “determine who hands over the box,” “verify the scoreboard change”). For complex query which requires multi-step reasoning, we prompt the PLANNER to first plan the initial observation and leave the following steps in the next rounds. By decomposing the complex queries, AVP achieves better handling in multi-hop reasoning and temporally dispersed evidence seeking.
- **where** is a targeted temporal region $[t_s, t_e]$. It is seeded from: (i) explicit timestamps in Q (e.g., “1:00–1:30”), (ii) soft textual cues (“opening scene,” “final minutes”). When no prior is available, we first sweep the entire video at low cost (low fps and spatial_res) to gather coarse evidence. Across rounds, this region can be tightened or shifted based on the Reflector’s feedback, enabling coarse-to-fine localization without dense scanning.
- **how** specifies sampling granularity for the long video observation, where $\text{how} = (\text{fps}, \text{spatial_res})$. The PLANNER determines the granularity of the targeted evidence and accordingly decides the sampling strategy. By default, it adopts coarse settings (lower fps and spatial_res) to perform low-cost exploration across the video and quickly identify potential evidence regions. When finer details are required—such as subtle object interactions or small spatial cues, the PLANNER increases sampling density to ensure more precise perception. This adaptive design allows AVP to allocate computation efficiently across granularities while maintaining high fidelity.

The resulting P serves as a compact, executable target observation that guides the Observer on what to looking for, which region of the video to inspect, and how to sample for efficient, query-focused observation.

3.2. Targeted Video Observation

Once the plan is generated, the observer (OBSERVER, a MLLM) executes the plan to gather detailed, time-stamped evidence from the video. Specifically, in round r , given the plan $P^{(r)} = (\text{what}^{(r)}, \text{where}^{(r)}, \text{how}^{(r)})$, the OBSERVER inputs the the query Q and instruction in $\text{what}^{(r)}$, the video segment defined by the temporal where and uses the sampling strategy in how (fps and spatial resolution). Instead of generating free-form text responses, the OBSERVER is prompted to produce structured, timestamp-aware evidence text in the form of

$$E \in \{([\text{start}_i, \text{end}_i], d_i)\}_{i=1}^N,$$

where each d_i is a concise, query-conditioned description of the visual event within the time interval $[\text{start}_i, \text{end}_i]$. Specifically, we maintain an evidence list $\mathcal{E}$ that accumulates evidence across rounds. At each round r , the OBSERVER generates new evidence E^r and append it to the cumulative

evidence list:

$$E^r = \text{OBSERVER}(V, Q, P^{(r)}), \quad \mathcal{E} \leftarrow \mathcal{E} \cup E^r.$$

This cumulative evidence list $\mathcal{E}$ serves as the working memory of AVP, allowing the reflector to assess sufficiency based on all past evidence and guiding the PLANNER’s subsequent updates. Compared with free-form captioning, this design yields more stable, query-relevant evidence and leads to better grounded reasoning over long videos. This targeted video observation design allows AVP to perceive only the most query-relevant portions of the video, keeping it efficient and avoiding redundant or irrelevant information.

3.3. Evidence Reflection and Re-Planning

After each observation round, AVP employs a reflector (REFLECTOR) to evaluate the sufficiency of the accumulated evidence and decide whether additional observation is required. The REFLECTOR verifies how well the collected evidence supports an answer, and when confidence is insufficient, it provides feedback for the next round of planning.

Evidence Reflection. At round r , given the query Q and the current cumulative evidence list $\mathcal{E}$, the REFLECTOR jointly produces a query confidence score $C^{(r)}$ and a justification $J^{(r)}$:

$$(C^{(r)}, J^{(r)}) = \text{REFLECTOR}(Q, \mathcal{E}),$$

where $C^{(r)} \in [0, 1]$ measures the confidence in evidence sufficiency to answer the given query, and $J^{(r)}$ specifies which answer the current evidence supports or what information is still missing. If the confidence is higher than the confidence threshold τ_{conf} , the REFLECTOR directly extracts the final answer from $J^{(r)}$; otherwise, the justification highlights missing or uncertain cues to guide the next round of planning step.

History Update and Re-Planning. When confidence remains below the threshold, the Reflector appends the current observation and justification to the running history H . The history provides the PLANNER with a concise summary of what has been inspected, verified, or left unresolved. The PLANNER then refines its next plan using this feedback:

$$P^{(r+1)} = \text{PLANNER.REPLAN}(Q, H, J^{(r)}),$$

shifting attention toward the regions, entities, or temporal spans identified as uncertain by the Reflector.

By iteratively running the plan-observe-reflect process, AVP forms a closed-loop perception–reasoning cycle that continuously refines its focus until the gathered evidence becomes sufficient. This iterative design allows the system to adaptively reason over long videos, reducing computation, and maintaining grounded, query-aligned understanding. We present full algorithm in Algorithm 1.

Algorithm 1: Active Video Perception(AVP)

Inputs : Video V , Query Q , Max Rounds $R_{\max}$, Confidence Threshold τ_{conf}

Output : Answer A , Justification J , Evidence List $\mathcal{E}$, History H

```
1  $P^{(1)} \leftarrow \text{PLANNER.INIT}(Q)$ ;  $\mathcal{E} \leftarrow []$ ;  $H \leftarrow []$  // Init plan, evidence list and history
2 for  $r \leftarrow 1$  to  $R_{\max}$  do
3    $E^{(r)} \leftarrow \text{OBSERVER}(V, Q, P^{(r)})$ 
4    $\mathcal{E} \leftarrow \mathcal{E} \cup E^{(r)}$  // accumulate this round's evidence
5    $(C^{(r)}, J^{(r)}) \leftarrow \text{REFLECTOR}(Q, \mathcal{E})$  // evidence reflection: confidence & justification
6   if  $C^{(r)} \geq \tau_{\text{conf}}$  then
7      $A \leftarrow \text{REFLECTOR.EXTRACTANSWER}(J^{(r)})$  // answer is entailed by justification
8     return  $A, J^{(r)}, \mathcal{E}, H$ 
9   if  $r = R_{\max}$  then
10     $A \leftarrow \text{REFLECTOR.FORCEANSWER}(Q, \mathcal{E})$  // force to give answer on final round
11    return  $A, J^{(r)}, \mathcal{E}, H$ 
12    $H \leftarrow H \cup \{(P^{(r)}, E^{(r)}, J^{(r)})\}$  // append plan & evidence & justification to history
13    $P^{(r+1)} \leftarrow \text{PLANNER.REPLAN}(Q, H, J^{(r)})$  // re-plan for additional observation
```

4. Experimental Setup

4.1. Datasets

We evaluate AVP on five diverse long video understanding benchmarks:

- (1) **MINERVA** [34] is a recent challenging video reasoning benchmark consisting of 1515 hand-crafted questions. The average video duration is 12 minutes.
- (2) **LV-Bench** [70] is a benchmark specifically designed for long video understanding which includes 1549 multiple-choice questions across 103 hour-long videos.
- (2) **MLVU** [84] is a multi-task Long Video Understanding Benchmark for the comprehensive and in-depth evaluation of LVU. We use the multiple-choice QA samples from the MLVU test split, containing 2175 video QA samples with more than 15 minutes average video duration.
- (4) **Video MME** [16] is a comprehensive evaluation benchmark for video analysis from short to long videos (average min for long split video). We use the standard split of Video-MME, which contains 2700 samples designed for both perception and reasoning tasks (900 samples with 41min average duration for the long split).
- (5) **LongVideoBench (LVB)** [68] is a video QA benchmark that highlights referred reasoning questions, which are dependent on long frame inputs. We test on the public validation split, which contains 1337 video reasoning questions (533 samples with 15-60 min video for long split).

4.2. Evaluation Metrics

We evaluate AVP under the multiple-choice QA setting. We use standard accuracy metrics for all experiments. We do not include auxiliary subtitle for all benchmarks.

4.3. Implementation Details

We adopt Gemini-2.5-Pro¹ [53] as our default MLLM agent for all components. We also provide the results with lightweight Gemini-2.5-Flash model in Tab. 1 and Tab. 4. We provided more ablation with different backbone models (including open-source models) in appendix. For fair comparison, we fix the max input token as 128K. If the input video (region) exceeds this budget, we uniformly sample the max frames that within the token limit. For spatial token setup (spatial_res), we follow Gemini’s MediaResolution setup to have 2 scale (low, medium, high), while low and medium is 66 and 258 tokens per frame, respectively. We set the max rounds $R_{\max}$ as 3 and confidence threshold τ_{conf} as 0.7. We provide additional analysis for the design choices in Sec. 5.2.2. We provided more implementation details (including detailed prompts) and analysis in appendix.

5. Results

5.1. Main Results on Long Video Benchmarks

Tab. 1 presents a comprehensive comparison of AVP against existing general-purpose MLLMs [35, 37, 53, 55], video-specific MLLMs [18, 45, 63, 67], and agentic video frameworks [8, 46, 62, 65, 79, 82, 88] across five video understanding benchmarks: MINERVA [34], LVBench [70], MLVU [84], Video-MME [16] and LongVideoBench [68].

Comparison with MLLMs. Among general-purpose multimodal LLMs, proprietary systems such as Gemini-2.5-Pro [53] and Seed-1.5-VL [55] achieve strong overall results but still fall short of our proposed AVP. In particular, AVP (w/

¹2025-06-17 version

Methods	MINERVA	LVBench	MLVU	Video-MME		LongVideoBench	
	Overall	Overall	Test	Overall	Long	Val	Long
<i>General-Purpose MLLMs</i>							
Seed-1.5-VL [55]	-	64.6	82.1	77.9	-	74.4	-
Qwen-3-VL [54]	-	67.7	84.3	79.2	-	-	-
GPT-4o [35]	45.5	48.9	54.9	71.9	65.3	66.7	60.9
GPT-4.1 [37]	54.0	63.4	-	72.0	-	-	-
Gemini-2.5-Flash [53]	54.6	56.7	72.4	74.2	69.1	66.2	61.8
Gemini-2.5-Pro [53]	<u>61.8</u>	67.4	79.6	<u>82.4</u>	77.6	69.8	66.6
<i>Video-Specific MLLMs</i>							
LongVU [45]	-	-	65.4	60.6	59.5	-	-
AdaReTaKe [63]	-	53.3	78.1	73.5	65.0	67.0	-
Video-RTS [67]	37.8	43.2	-	63.0	54.1	56.6	52.2
FrameMind [18]	-	-	48.6	60.9	57.5	-	-
<i>Agentic Video Frameworks</i>							
VideoAgent [62]	-	29.3	64.4	-	46.4	-	-
VideoTree [65]	40.2	28.8	60.4	60.6	54.2	-	-
SiLVR [79]	44.4	-	45.2	74.1	<u>77.7</u>	-	-
VideoLucy [88]	-	58.8	76.1	72.5	66.8	-	-
Vgent [46]	-	-	72.1	68.9	-	59.7	-
LVAgent [8]	-	-	83.9	<u>81.7</u>	74.3	80.0	-
DeepVideoDiscovery (DVD) [82]	-	<u>74.2</u>	-	-	67.3	<u>71.6</u>	<u>68.6</u>
<i>Active Video Perception (Ours)</i>							
AVP w Gemini-2.5-Flash	56.9 (+2.3)	63.8 (+7.1)	74.1 (+1.7)	81.2 (+7.0)	76.7 (+7.6)	70.2 (+4.0)	65.5 (+3.7)
AVP w Gemini-2.5-Pro	65.6 (+3.8)	74.8 (+7.4)	84.3 (+4.7)	85.3 (+2.9)	81.9 (+4.3)	73.4 (+3.6)	70.0 (+3.4)

Table 1. Comparison with general-purpose MLLMs, Video-specific MLLMs, and agentic video frameworks on five long video understanding benchmarks (MINERVA, LVBench, MLVU, Video-MME, LongVideoBench). We **bold** the best and underline the second-best result in each column. Results shows that AVP achieves best overall accuracy on all datasets across different baselines, achieving significant improvements on its backbone model (in **blue**) across all benchmark. We **gray** out the results that use auxiliary subtitle information.

Gemini-2.5-Pro) surpasses the state-of-the-art Gemini-2.5-Pro model [53] by **4.5%** average accuracy over all benchmarks, demonstrating that direct inference over full length remains insufficient for complex, long-horizon queries that require targeted evidence seeking. AVP (w/ Gemini-2.5-Flash) also outperforms its backbone by **4.4%**, showing generalization ability of the proposed framework in weaker backbone MLLMs. Meanwhile, AVP significantly outperforms the video-specific MLLMs, including compression-based methods [45, 63] and (visual) Chain-of-Thoughts methods [18, 67]. This result highlights the active perception concept for long video understanding and encourages future research.

Comparison with Agentic Frameworks. Within the class of agentic video reasoning systems, AVP consistently achieves the best (or second-best) results across all benchmarks. We compare AVP with six recent agentic video frameworks, including VideoAgent [62], VideoTree [65], SiLVR [79], VideoLucy [88], LVAgent [8] and DeepVideoDiscovery (DVD) [82]. We find that AVP achieves best performance against all baseline methods and signifi-

cant improvement compared to the backbone model in all benchmarks. Comparing to the recent VideoLucy and DVD methods, AVP achieves **10.5%** and **5.7%** average improvements while both using strong LLM backbones (DeepSeek-R1 [10] for VideoLucy, and OpenAI-o3 [38] for DVD). We also compared the efficiency in term of inference time with DVD in Tab. 2, showing AVP is not only more performant, but also significantly efficient. These results validate the effectiveness of active perception for long video understanding : rather than passively encoding frames, AVP plans what to observe, observes purposefully, and reflects adaptively, leading to higher accuracy and greater efficiency than both MLLMs and recent agentic frameworks.

5.2. Quantitative Analysis

In this section, we analyze different aspect of AVP, including efficiency analysis, ablation study on AVP’s design choices. We provided more quantitative analysis in the appendix.

Method	Avg. Inference Time (s)	Avg. Input Tokens (K)	Acc
DVD	790.5	1071.6	74.2
AVP (Ours)	145.3	132.5	74.8

Table 2. Efficiency comparison on LVBench. We report average inference time in seconds, average input token count, and accuracy. By actively querying the video rather than passively captioning all clips, AVP achieves better overall efficiency and accuracy.

Method	MINERVA	LVBench
Observer (Baseline)	60.8	67.4
Planner + Observer	63.9	72.6
Planner + Observer + Reflector (AVP)	65.6	74.8

Table 3. **Component ablation of AVP.** Adding the Planner and then the Reflector on top of the Observer baseline consistently improves MINERVA and LVBench accuracy, showing that query-conditioned planning and reflection are key to AVP’s performance.

5.2.1. Efficiency Analysis

As shown in Tab. 2, we evaluate inference efficiency on LVBench in terms of average runtime, average input token count, and accuracy. DVD [82] requires 790.5s per video and processes on average 1.07M tokens. Notably, a finer breakdown shows that its captioning stage alone takes 637.2s and consumes roughly 0.9M tokens. In contrast, AVP eliminates this query-agnostic captioning stage and performs only targeted query reasoning, reducing inference time to 145.3s, achieving $5.44\times$ faster (**81.6%** reduction). Meanwhile, AVP only consumes **12.4%** of the input tokens compared to DVD while improving the LVBench accuracy. These results indicate that **actively** deciding what, where, and how to observe not only removes redundant caption processing but also strengthens reasoning by concentrating computation on query-relevant content.

5.2.2. Ablation Study

AVP Components. We conduct a step-wise ablation to assess the contribution of each component in AVP. As shown in Tab. 3, introducing the PLANNER notably improves both MINERVA and LVBench accuracy, demonstrating the benefit of query-conditioned multi-step exploration over static observation. The PLANNER guides the agent to allocate computation toward potentially informative regions rather than processing frames uniformly. Adding the REFLECTOR yields a further performance gain, confirming that iterative process enhances reasoning trustworthy. Together, these results highlight that active perception, planning what to observe and reflecting on what has been seen substantially strengthens long video understanding.

Model Selection. Table 4 examines the impact of varying the model selection across Planner, OBSERVER, and RE-

PLANNER	OBSERVER	REFLECTOR	MINERVA	LVBench
2.5-Flash	2.5-Flash	2.5-Flash	56.9	63.8
2.5-Pro	2.5-Flash	2.5-Pro	60.2	67.6
2.5-Flash	2.5-Pro	2.5-Flash	63.6	71.8
2.5-Pro	2.5-Pro	2.5-Pro	65.6	74.8

Table 4. **Agent MLLM selection within AVP.** We vary Gemini-2.5 Flash/Pro backbones for the PLANNER, OBSERVER, and REFLECTOR, stronger components consistently improve performance on both benchmarks.

Max Rounds	MINERVA	LVBench
1	63.9	72.6
2	65.0	74.6
3	65.6	74.8
5	65.5	74.6

Table 5. **Ablation on max round limit.** Increasing the number of max round limit improves performance on both benchmarks and gets best results by three rounds, indicating that only a few interaction steps are sufficient.

FLECTOR within AVP under Gemini-2.5 [53] family (we add additional model ablation in supp.). We observe that both benchmarks benefit from stronger components, but their sensitivities differ. On MINERVA, which features complex, multi-hop reasoning queries, performance improves substantially with stronger **Planner** and **Reflector** models, indicating that strategic planning and reflective consolidation are crucial for handling compositional reasoning. In contrast, LVBench, characterized by extremely long videos, relies more heavily on a robust **Observer**, the component directly responsible for navigating and gathering evidence efficiently from vast temporal spans. The best configuration employs powerful models across all three modules, confirming that AVP’s active perception design yields synergistic gains in both reasoning depth and temporal scalability.

Max Round Limit. Table 5 studies how the number of Plan–Observe–Reflect rounds affects performance. Both MINERVA and LVBench show steady gains from one to three rounds, confirming that iterative reasoning enables AVP to progressively refine its evidence set and improve decision confidence. The improvement is more pronounced on MINERVA, where multi-hop reasoning benefits from repeated reflection and targeted re-observation. Beyond three rounds, performance saturates, suggesting that AVP has already acquired sufficient evidence and additional cycles bring limited benefit. This result validates the efficiency of our design, AVP achieves strong reasoning capability with only a few lightweight interaction rounds.

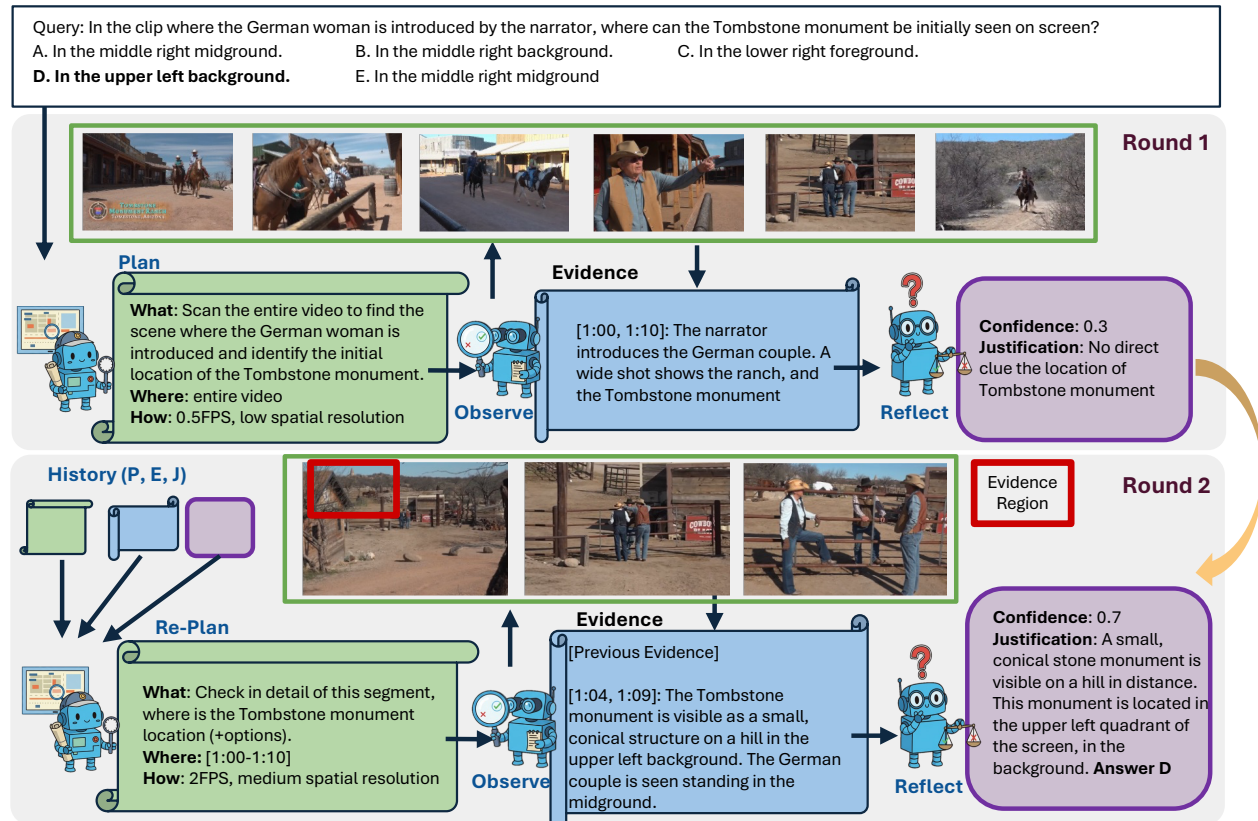


Figure 3. **Qualitative example of AVP.** Given a multiple-choice query about the Tombstone monument’s first on-screen appearance, Round 1 performs a coarse scan of the entire video (0.5 FPS, low resolution) and localizes a candidate interval [1:00, 1:10], but the REFLECTOR judges the evidence insufficient. Round 2 re-plans a targeted pass over this window (2 FPS, medium resolution), enabling the OBSERVER to localize the monument in the upper-left background and the REFLECTOR to confidently select the correct answer (option D) and halt.

5.3. Visualization

In Fig. 3, we illustrate how AVP acquires and verifies evidence through a multi-round Plan–Observe–Reflect loop on a long video. Given the query, “In the clip where the German woman is introduced by the narrator, where can the Tombstone monument be initially seen on screen?”, Round 1 uses a coarse, uniform sweep to localize candidate moments (0.5 FPS, low resolution). This pass narrows the search to the [1:00, 1:10] interval but the reflector flags the observations as insufficient due to lack of detail, prompting a refined follow-up (Round 2). In Round 2, the planner schedules a targeted revisit over [1:00, 1:10] at 2 FPS with medium resolution, and the observer extracts query-relevant cues: the Tombstone monument appears as a small, conical structure on a hill in the upper-left background while the German couple stands in the mid-ground. The evidence list is now sufficient for the reflector to stop and produce the final answer, demonstrating AVP’s coarse-to-fine scheduling, evidence-grounded verification. We provided additional visualization samples with different scenario (start with a grounded video region from query prior and refine the region based on the observation in

the next round) and failure case in appendix.

6. Conclusion

Inspired by active perception theory, we present **Active Video Perception (AVP)**, which handles long video understanding as an iterative, query-driven evidence seeking process. Rather than passively caption the video frames, AVP treats the video as an interactive environment and actively decides what to inspect, where to focus, and at what granularity in order to acquire compact, time-stamped evidence directly from pixels. Concretely, AVP runs an iterative plan–observe–reflect process using MLLM agents. Empirically, AVP achieves best overall accuracy among agentic frameworks across five long video benchmarks, and surpasses the leading agentic method (DVD) by **5.7%** in average accuracy while only requiring **18.4%** inference time and **12.4%** input tokens. Our ablation study shows that AVP achieves significant improvement under different MLLM backbones, validating the robustness. Looking ahead, an exciting direction is extending active video perception to embodied agents that must decide what and when to observe while acting under real-world physical constraints.

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